PROJECT GUIDE

Customer Churn Analytics & Risk Scoring API

This is a self-directed case study, built using a public 1,000,000-row dataset to demonstrate the full process I'd apply to a real customer base: from raw data to a validated model to a deployed, working tool. It has three parts, each usable on its own, but designed to be read together.

1 Interactive Dashboard

A tiered walkthrough of the analysis itself: data cleaning, customer segmentation, churn drivers, survival modelling and two live tools you can adjust yourself, a customer risk calculator and a retention-offer simulator. Opens directly in any browser, no setup required.

2 Decision Report (PDF / Word)

The same analysis written up as a formal business document: an executive summary, a three-scenario cost comparison, honestly stated model limitations and a concrete recommended next step. This is the format a real client engagement would produce.

3 Live API + Risk Checker

The modelling isn't just theoretical here, it's deployed. The API is live and publicly callable. The linked page sends real requests to it and shows the real response. Interactive documentation for the API itself is available at the link below.

risk-scoring-api-1tct.onrender.com/docs

A note on the live demo

The API runs on a free hosting tier, which spins down after 15 minutes of inactivity. If it's been quiet, the first request can take 30–50 seconds to respond while it wakes back up, that's expected, not a fault.

What this demonstrates

- Data cleaning, exploratory analysis and statistical testing
- Unsupervised segmentation and survival modelling (Kaplan-Meier, Cox proportional hazards)
- Predictive modelling with proper held-out validation (AUC, calibration, precision/recall)
- Treatment-effect (uplift) modelling and budget-constrained optimisation
- A deployed, publicly callable API (FastAPI, Docker, Render)
- Clear written communication for a non-technical business audience

A short walkthrough video is in progress and will be linked here once ready. In the meantime, all three pieces above are fully explorable on their own.

Tamearon Mullings · tamearonmullings@gmail.com